

Date	01 October 2026
Team ID	SWUID2026355656
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Project Design Phase-II

Technology Stack (Architecture & Stack)

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

[Insert Architectural Diagram Here - System Flow]

Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	React, FastAPI, PostgreSQL

2.	Security Implementations	Access controls, authentication, encryption	nJWT, HTTPS, bcrypt, RBAC
3.	Scalable Architecture	Justify the scalability (Microservices)	Docker Containers, Kubernetes
4.	Availability	Use of load balancers, distributed servers	AWS Auto Scaling
5.	Performance	Design consideration for caching, CDNs	Redis, Cloudflare CDN

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application (Web UI)	HTML, CSS, React.js
2.	Application Logic-1	Backend API logic and routing	Python / FastAPI
3.	Application Logic-2	Authentication & Authorization logic	JWT / OAuth2
4.	Application Logic-3	Email & Notifications process	SendGrid / Celery
5.	Database	Relational data type (Users, Books)	PostgreSQL
6.	Cloud Database	Database Service on Cloud	Amazon RDS
7.	File Storage	File storage for covers/avatars	AWS S3
8.	External API-1	Fetching Book metadata API	Google Books API
9.	External API-2	Payment processing API	Stripe API
10.	Machine Learning	Book recommendation Engine	Scikit-Learn
11.	Infrastructure	Application Deployment (Cloud)	AWS EC2, Kubernetes

Table-2: Application Characteristics:

References:

<https://c4model.com/> <https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/> <https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>